

Backend Coding Challenge

Introduction: This challenge focuses on building a core data synchronization and reporting system for a digital bookstore. You are tasked with implementing a CSV-based inventory upsert API and a PDF report generator that lists the top 5 books by price and top 5 authors by book count.

Requirements:

- **Tech Stack:** Node.js and Express.
- **Database Schema:**
 - **store:** `id, name, address, logo`
 - **book:** `id, name, pages, author_id`
 - **author:** `id, name`
 - **store_book:** (many-to-many relationship) `store_id, book_id, price, copies, sold_out(flag)`
- **ORM:** Preferably Sequelize, but any other ORM is acceptable.
- **Endpoints:**
 - An endpoint to accept a CSV file, parse it, and ingest the data.
 - **POST** `/api/inventory/upload`
 - CSV Format: `store_name, store_address, book_name, pages, author_name, price, logo`
 - Process each row by attempting to **match** it to an existing store, book, and author; otherwise, **create** the new entities. If the store already stocks the specific book, **update** the record by incrementing the number of copies.
 - An endpoint to generate a store report
 - **GET** `/api/store/:id/download-report`
 - PDF report should be named `[Store Name]-Report-YYYY-MM-DD.pdf`.
 - The PDF report should have the store logo and name as well as the:
 - Top **5** Priciest Books: The **5** books with the highest prices.
 - Top **5** Prolific Authors: The **5** authors with the greatest number of available books in the inventory.
- Implement necessary validation and robust error handling for both endpoints.
- **Containerization:** The backend should be containerized using Docker (nice to have).
- **Orchestration:** Use Docker Compose to start the backend (nice to have).

Deliverables:

- You are required to submit this exercise within **3 hours** of receiving it. Please report the actual time spent on delivering these requirements.
- Submit the code as a public repository on any well-known platform such as GitHub or Bitbucket. Once the code is pushed to the repo and it is ready, reply back to the email with the repository link.
- Include a README in the repository explaining how to get the containers up and running, or how to start the app.